

▢ Fermat's Theorem : (For extrema)

Suppose that $a < c < b$. If a function f is defined on the interval (a, b) , and it has a maximum or minimum at c , then either f' doesn't exist at c or $f'(c) = 0$.

Proof: let $a < c < b$ and the function is ~~no~~ f is maximum at c .

$$\Rightarrow \cancel{f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}} \quad x = c+h$$

Thus $f(c) \geq f(x) \quad \forall x$ for all x sufficiently close to c .

We have $f(c) \geq f(c+h)$ - say h is very close to 0.

$$\Rightarrow 0 \geq f(c+h) - f(c)$$

If $h > 0$, then

$$0 \geq \frac{f(c+h) - f(c)}{h} \\ \Rightarrow 0 \geq \lim_{h \rightarrow 0^+} \frac{f(c+h) - f(c)}{h} \quad \text{--- (i)}$$

If $h < 0$, then

$$0 \leq \frac{f(c+h) - f(c)}{h} \\ \Rightarrow 0 \leq \lim_{h \rightarrow 0^-} \frac{f(c+h) - f(c)}{h} \quad \text{--- (ii)}$$

From eqs (i) & (ii), we conclude that

$$\lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} = 0$$

$$\text{i.e. } f'(c) = 0$$

since f is differentiable at $x=c$.

▢

Rolle's Theorem:

If f is a continuous function on $[a, b]$ and differentiable on (a, b) , with $f(a) = f(b) = 0$, then there exists some c in (a, b) where $f'(c) = 0$.

Proof: Case I:

If $f(x) = 0 \quad \forall x \in [a, b]$, then nothing to prove.

Case II:

If $f(x) \neq 0$ for some $x \in (a, b)$.

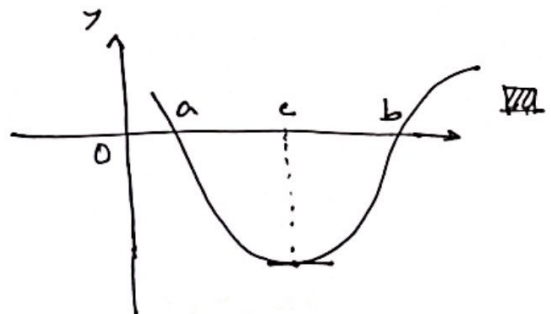
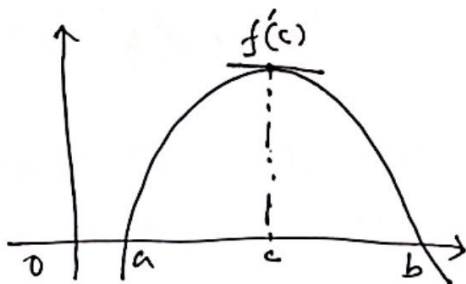
Assume that $f(x) > 0$ at some point in (a, b) . Since f is continuous on $[a, b]$.

By extreme-value theorem, f has an absolute maximum on $[a, b]$. Since $f(a) = f(b) = 0$.

By Fermat's theorem, the absolute maximum must occur at some point c in (a, b) . It follows that c is a critical point of f , and since f is differentiable on (a, b) which implies that $f'(c) = 0$.

Similarly, assume that $f(x) < 0$ at some point in (a, b) .

Similar way we conclude that $f'(c) = 0$.



The Mean-Value Theorem:

Let f be continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) . Then there is at least one point c in (a, b) such that

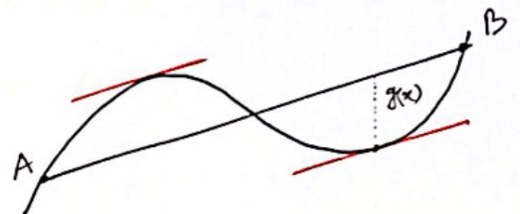
$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof: Suppose f satisfies the hypothesis of the Mean Value Th^m.

The line joining the points $(a, f(a))$ and $(b, f(b))$ has eqⁿ:

$$y = f(a) + \frac{f(b) - f(a)}{b - a} (x - a) \quad \text{--- ①}$$

Let g be the difference betⁿ f and the line ①



That is

$$g(x) = f(x) - \left[f(a) + \frac{f(b) - f(a)}{b - a} (x - a) \right]$$

Since g is the difference of two continuous functions. So g is continuous on $[a, b]$.

Again g is the difference of two differentiable functions. So g is differentiable on (a, b) .

$$\text{So, } g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}$$

Moreover, $g(a) = 0$, and $g(b) = 0$

By Rolle's theorem, $\exists c \in (a, b)$ with $g'(c) = 0$

$$\begin{aligned}
 g'(c) &= 0 \\
 \Rightarrow f'(c) - \frac{f(b) - f(a)}{b - a} &= 0 \\
 \Rightarrow f'(c) &= \frac{f(b) - f(a)}{b - a}
 \end{aligned}$$

□

Problem: Let f and g be functions, continuous on $[a, b]$, differentiable on (a, b) and let $f(a) = f(b) = 0$. Prove that there is a point $c \in (a, b)$ such that $g'(c)f(c) + f'(c) = 0$.

Sol. We define $h(x) = f(x)e^{g(x)}$.

Since f and g continuous and differentiable. So, h is also continuous on $[a, b]$ and differentiable on (a, b) .

$$h(a) = f(a)e^{g(a)} = 0 = h(b)$$

By Rolle's Th^m, $\exists c \in (a, b)$ such that

$$h'(c) = 0$$

$$\begin{aligned}
 \text{Since } h'(x) &\Rightarrow f'(x)e^{g(x)} + f(x)e^{g(x)}g'(x) \\
 &= [f'(x) + g'(x)f(x)]e^{g(x)}
 \end{aligned}$$

$$\begin{aligned}
 \therefore h'(c) &= [f'(c) + g'(c)f(c)]e^{g(c)} = 0, \quad e^{g(c)} \neq 0 \\
 \Rightarrow f'(c) + g'(c)f(c) &= 0
 \end{aligned}$$

* $\square$

Problem: Suppose that f is continuous and differentiable on $[0, 1]$, that $f(x)$ is in $[0, 1]$ for each x , and that $f'(x) \neq 1$ for all x in $[0, 1]$. Show that there is EXACTLY ONE number x in $[0, 1]$ such that $f(x) = x$. Note that there are two things to show: existence and uniqueness.

Sol. Here $f'(x) \neq 1$ and $f(x) = x \quad \exists$ one x .

We want to show the existence and uniqueness for this x .

$$\text{Say } g(x) = f(x) - x$$

If $f(0) = 0$ and $f(1) = 1$ then the statement is proved.

Now say, $f(0) \neq 0$ and $f(1) \neq 1$

Consider $f(0) > 0$, $f(1) < 1$

$$\text{Then } g(0) = f(0) - 0 > 0, \quad g(1) = f(1) - 1 < 0$$

Base on IVT, there exist at least one point such that

$$g(x) = 0 \Rightarrow f(x) = x. \quad [\text{existence}]$$

For uniqueness:

Say $f(a) = a$, $f(b) = b$ where $a \neq b$

By the Mean Value Th^m, $\exists c \in (a, b)$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{b - a}{b - a} = 1$$

But $f'(x) \neq 1$. So which is contradiction.

That means $a = b$.

So, $f(x) = x$ has exactly one solution.

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Problem: Suppose a differentiable function f satisfies $|f'(x)| \leq 1$ for all $x \in (0, 1)$. If $f(x_0) \geq 1$ for some $x_0 \in (0, 1)$, show that f is positive on the whole interval i.e. $f(x) > 0$ for all $x \in (0, 1)$.

Sol. let's take $x \in (0, 1)$ and $x > x_0$. Apply the MVT,

$$\frac{f(x) - f(x_0)}{x - x_0} = f'(c) \quad \text{where } x_0 < c < x.$$

since $|f'(c)| \leq 1$ so

$$\left| \frac{f(x) - f(x_0)}{x - x_0} \right| \leq 1$$

$$\frac{|f(x) - f(x_0)|}{|x - x_0|} \leq 1$$

$$|f(x) - f(x_0)| \leq |x - x_0| < 1, \quad \text{where } x \in (0, 1) \text{ and } x > x_0.$$

$$|f(x) - f(x_0)| < 1$$

$$-1 < f(x) - f(x_0) < 1$$

$$f(x_0) - 1 < f(x) < f(x_0) + 1$$

$$\text{so, } f(x) > f(x_0) - 1$$

since, $f(x_0) \geq 1$ which implies $f(x) > 0$ for all $x \in (0, 1)$.

✱

Extra Problems:

1. Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable functions satisfying $f'(x) > g'(x)$ for all $x \in \mathbb{R}$ and $f(0) = g(0)$. Prove that $f(x) > g(x)$ for $x > 0$ and $f(x) < g(x)$ for $x < 0$.
2. Suppose that f is continuous on the closed interval $[a, b]$, that f is differentiable on the open interval (a, b) , and that $f'(x) = 0$ for $x \in (a, b)$. Then f is constant on that interval $[a, b]$.
3. Let $f: [0, \infty) \rightarrow \mathbb{R}$ be a continuous function such that $f(0) = 0$. If f is differentiable on $(0, \infty)$ and f' is increasing on $(0, \infty)$ then show that g is an increasing function where g defined as $g: (0, \infty) \rightarrow \mathbb{R}$ by $g(x) = \frac{f(x)}{x}$.